高等线性代数选讲期末基本计算练习参考答案

1. LU 分解与线性方程组

$$A=\begin{pmatrix}1 & 2 & -1 & 1\\ 2 & 5 & 1 & 0\\ -1 & 1 & 12 & -6\\ 1 & 0 & -3 & 10\end{pmatrix}.$$

可取

$$L=\begin{pmatrix}1 & 0 & 0 & 0\\ 2 & 1 & 0 & 0\\ -1 & 3 & 1 & 0\\ 1 & -2 & 2 & 1\end{pmatrix},\qquad U=\begin{pmatrix}1 & 2 & -1 & 1\\ 0 & 1 & 3 & -2\\ 0 & 0 & 2 & 1\\ 0 & 0 & 0 & 3\end{pmatrix}.$$

直接相乘可验 $A=LU$。

求解 $Ax=b$ 时，先解 $Ly=b$：

$$\begin{cases}y_1=-3,\\ 2y_1+y_2=-1,\\ -y_1+3y_2+y_3=22,\\ y_1-2y_2+2y_3+y_4=-5,\end{cases}\quad\Longrightarrow\quad y=\begin{pmatrix}-3\\ 5\\ 4\\ 0\end{pmatrix}.$$

再解 $Ux=y$：

$$\begin{cases}3x_4=0,\\ 2x_3+x_4=4,\\ x_2+3x_3-2x_4=5,\\ x_1+2x_2-x_3+x_4=-3,\end{cases}\quad\Longrightarrow\quad x=\begin{pmatrix}1\\ -1\\ 2\\ 0\end{pmatrix}.$$

2. Cholesky 分解

$$A=\begin{pmatrix}4&2&0&2\\2&5&2&1\\0&2&5&2\\2&1&2&3\end{pmatrix}.$$

可取

$$L=\begin{pmatrix}2&0&0&0\\1&2&0&0\\0&1&2&0\\1&0&1&1\end{pmatrix},$$

因为

$$LL^T=\begin{pmatrix}4&2&0&2\\2&5&2&1\\0&2&5&2\\2&1&2&3\end{pmatrix}=A.$$

先解 $Ly=b$：

$$\begin{cases}2y_1=8,\\ y_1+2y_2=6,\\ y_2+2y_3=9,\\ y_1+y_3+y_4=10,\end{cases}\quad\Longrightarrow\quad y=\begin{pmatrix}4\\1\\4\\2\end{pmatrix}.$$

再解 $L^Tx=y$：

$$\begin{cases}x_4=2,\\ 2x_3+x_4=4,\\ 2x_2+x_3=1,\\ 2x_1+x_2+x_4=4,\end{cases}\quad\Longrightarrow\quad x=\begin{pmatrix}1\\0\\1\\2\end{pmatrix}.$$

3. QR 分解与最小二乘

$$A=\begin{pmatrix}1&1&0\\1&0&1\\0&1&1\\1&-1&1\end{pmatrix},\qquad a_1=\begin{pmatrix}1\\1\\0\\1\end{pmatrix},\ a_2=\begin{pmatrix}1\\0\\1\\-1\end{pmatrix},\ a_3=\begin{pmatrix}0\\1\\1\\1\end{pmatrix}.$$

先取

$$q_1 = \frac{a_1}{\|a_1\|} = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 0 \\ 1 \end{pmatrix}.$$

又因为 $a_1^T a_2 = 0$，所以

$$q_2 = \frac{a_2}{\|a_2\|} = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 0 \\ 1 \\ -1 \end{pmatrix}.$$

第三列做正交化：

$$u_3 = a_3 - (q_1^T a_3) q_1 - (q_2^T a_3) q_2.$$

其中

$$q_1^T a_3 = \frac{2}{\sqrt{3}}, \qquad q_2^T a_3 = 0,$$

故

$$u_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix} - \frac{2}{3} \begin{pmatrix} 1 \\ 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -\frac{2}{3} \\ \frac{1}{3} \\ 1 \\ \frac{1}{3} \end{pmatrix}, \qquad \|u_3\| = \frac{\sqrt{15}}{3}.$$

$$q_3 = \frac{u_3}{\|u_3\|} = \frac{1}{\sqrt{15}} \begin{pmatrix} -2 \\ 1 \\ 3 \\ 1 \end{pmatrix}.$$

于是

$$Q = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & -\frac{2}{\sqrt{15}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{1}{\sqrt{15}} \\ 0 & \frac{1}{\sqrt{3}} & \frac{3}{\sqrt{15}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{15}} \end{pmatrix},$$

$$R = \begin{pmatrix} \sqrt{3} & 0 & \frac{2}{\sqrt{3}} \\ 0 & \sqrt{3} & 0 \\ 0 & 0 & \frac{\sqrt{15}}{3} \end{pmatrix}.$$

最小二乘解满足

$$R\hat{x} = Q^T b.$$

先算

$$Q^T b = \begin{pmatrix} \frac{5}{\sqrt{3}} \\ 0 \\ \frac{\sqrt{15}}{3} \end{pmatrix}.$$

解得

$$\hat{x} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

于是

$$A\hat{x} = \begin{pmatrix} 2 \\ 2 \\ 2 \\ 1 \end{pmatrix}, \qquad r = b - A\hat{x} = \begin{pmatrix} 1 \\ -3 \\ 1 \\ 2 \end{pmatrix}.$$

$$\|r\|_2 = \sqrt{15}.$$

4. 对称矩阵的正交对角化

$$S = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix} = I + \mathbf{1}\mathbf{1}^T.$$

特征值为

$$\lambda_1 = 4, \qquad \lambda_2 = \lambda_3 = 1.$$

可取一组标准正交特征向量

$$u_1 = \frac{1}{\sqrt{3}}\begin{pmatrix}1\\1\\1\end{pmatrix}, \quad u_2 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\\0\end{pmatrix}, \quad u_3 = \frac{1}{\sqrt{6}}\begin{pmatrix}1\\1\\-2\end{pmatrix}.$$

令

$$Q = (u_1, u_2, u_3), \qquad D = \operatorname{diag}(4, 1, 1),$$

则

$$S = QDQ^T.$$

Rayleigh 商的最大值与最小值分别等于最大、最小特征值，即

$$\max R(x) = 4, \qquad \min R(x) = 1.$$

并且最大值在 u_1 方向取得，最小值在 $\operatorname{span}\{u_2, u_3\}$ 中取得。

由

$$S^{-1} = QD^{-1}Q^T$$

得

$$S^{-1} = \begin{pmatrix} \frac{3}{4} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{3}{4} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} \end{pmatrix}.$$

5. Jordan 标准形与矩阵幂

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -2 \end{pmatrix}.$$

该矩阵已经是 Jordan 形，即

$$J = \operatorname{diag}\left(\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, (-2) \right).$$

极小多项式为

$$m_A(\lambda) = (\lambda - 1)^3 (\lambda + 2).$$

记

$$J_3 = I + N, \qquad N = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad N^3 = 0.$$

则

$$J_3^n = (I+N)^n = I + nN + \frac{n(n-1)}{2}N^2.$$

故

$$A^n = \begin{pmatrix} 1 & n & \frac{n(n-1)}{2} & 0 \\ 0 & 1 & n & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & (-2)^n \end{pmatrix}.$$

6. 矩阵指数与线性微分方程

$$A = 2I + N, \qquad N = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \qquad N^3 = 0.$$

因此

$$e^{tA} = e^{2t}e^{tN} = e^{2t}\left(I + tN + \frac{t^2}{2}N^2\right).$$

而

$$N^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

所以

$$e^{tA} = e^{2t}\begin{pmatrix} 1 & t & \frac{t^2}{2} \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}.$$

于是

$$X(t) = e^{tA}X(0) = e^{2t}\begin{pmatrix} 1 & t & \frac{t^2}{2} \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = e^{2t}\begin{pmatrix} 1+\frac{t^2}{2} \\ t \\ 1 \end{pmatrix}.$$

7. 奇异值分解

$$A = \begin{pmatrix} 2 & 2 \\ 1 & -1 \\ 1 & -1 \end{pmatrix}.$$

先算

$$A^TA = \begin{pmatrix} 6 & 2 \\ 2 & 6 \end{pmatrix}.$$

其特征多项式为

$$(\lambda - 8)(\lambda - 4) = 0,$$

故特征值为 $8, 4$，奇异值为

$$\sigma_1 = 2\sqrt{2}, \qquad \sigma_2 = 2.$$

对应的单位特征向量可取为

$$v_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad v_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

于是

$$u_1 = \frac{Av_1}{\sigma_1} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad u_2 = \frac{Av_2}{\sigma_2} = \begin{pmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}.$$

再补一个与 u_1, u_2 正交的单位向量

$$u_3 = \begin{pmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}.$$

因此可取

$$U=\begin{pmatrix}1 & 0 & 0\\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}}\end{pmatrix},\quad \Sigma=\begin{pmatrix}2\sqrt{2} & 0\\ 0 & 2\\ 0 & 0\end{pmatrix},\quad V=\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}.$$

从而

$$A=U\Sigma V^{T}.$$

8. 二次多项式最小二乘拟合

法方程写成

$$\begin{pmatrix}\sum x_i^4 & \sum x_i^3 & \sum x_i^2\\ \sum x_i^3 & \sum x_i^2 & \sum x_i\\ \sum x_i^2 & \sum x_i & 4\end{pmatrix}\begin{pmatrix}a\\ b\\ c\end{pmatrix}=\begin{pmatrix}\sum x_i^2 y_i\\ \sum x_i y_i\\ \sum y_i\end{pmatrix}.$$

对数据

$$(-1,1),\ (0,0),\ (1,2),\ (2,5)$$

有

$$\sum x_i=2,\ \sum x_i^2=6,\ \sum x_i^3=8,\ \sum x_i^4=18,$$

$$\sum y_i=8,\ \sum x_i y_i=11,\ \sum x_i^2 y_i=23.$$

故法方程为

$$\begin{pmatrix}18 & 8 & 6\\ 8 & 6 & 2\\ 6 & 2 & 4\end{pmatrix}\begin{pmatrix}a\\ b\\ c\end{pmatrix}=\begin{pmatrix}23\\ 11\\ 8\end{pmatrix}.$$

解得

$$a=1,\qquad b=\frac{2}{5},\qquad c=\frac{3}{10}.$$

所以最小二乘拟合多项式为

$$y=x^2+\frac{2}{5}x+\frac{3}{10}.$$

对应残差为

$$r = \begin{pmatrix} \frac{1}{10} \\ -\frac{3}{10} \\ \frac{3}{10} \\ -\frac{1}{10} \end{pmatrix},$$

故最小残差平方和为

$$\|r\|_2^2 = \frac{1}{100} + \frac{9}{100} + \frac{9}{100} + \frac{1}{100} = \frac{1}{5}.$$

9. 矩阵函数计算

$$B = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix} = \begin{pmatrix} J & 0 \\ 0 & 2 \end{pmatrix}, \qquad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

先有

$$J^2 = -I_2, \qquad B^2 = \begin{pmatrix} -I_2 & 0 \\ 0 & 4 \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 4 \end{pmatrix}.$$

由分块计算

$$e^{tB} = \begin{pmatrix} e^{tJ} & 0 \\ 0 & e^{2t} \end{pmatrix}.$$

而

$$e^{tJ} = \cos t\, I_2 + \sin t\, J = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}.$$

故

$$e^{tB} = \begin{pmatrix} \cos t & -\sin t & 0 \\ \sin t & \cos t & 0 \\ 0 & 0 & e^{2t} \end{pmatrix}.$$

同理

$$\sin B = \begin{pmatrix} \sin J & 0 \\ 0 & \sin 2 \end{pmatrix}, \qquad \cos B = \begin{pmatrix} \cos J & 0 \\ 0 & \cos 2 \end{pmatrix}.$$

又因为

$$\sin J = \sinh(1)\, J, \qquad \cos J = \cosh(1)\, I_2,$$

所以

$$\sin B = \begin{pmatrix} 0 & -\sinh 1 & 0 \\ \sinh 1 & 0 & 0 \\ 0 & 0 & \sin 2 \end{pmatrix},$$

$$\cos B = \begin{pmatrix} \cosh 1 & 0 & 0 \\ 0 & \cosh 1 & 0 \\ 0 & 0 & \cos 2 \end{pmatrix}.$$